

XIANGLIN HUANG

CURRICULUM VITAE

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Position: Visiting Research Professor, Dongguk University [ORCID](#) [impacthuang.github.io](https://github.com/impacthuang)

Overview: I am a postdoctoral researcher at KAIST and a Visiting Research Professor at Dongguk University, specializing in the impact safety and prognostics & health management (PHM) of lithium-ion batteries, and in data-driven constitutive modeling of metals under impact loading. At the Dongguk iPHM Lab, I lead an independent research stream on lithium-ion battery impact safety, co-supervising graduate students on experimental failure-mode characterization, deep-learning frameworks for acoustic-emission-based damage detection, and multiphysics finite-element simulation of battery packs under dynamic loading.

In parallel, I specialize in data-driven constitutive modeling of metals under impact loading. A four-paper series I authored in the International Journal of Impact Engineering (2023–2027) establishes a complete framework, combining SVD/CP tensor decomposition with artificial neural networks, for verifying, determining, and validly extrapolating dynamic flow stress equations from discrete SHPB and Taylor-Hopkinson experimental data.

I have authored 18 SCIE-indexed journal papers (16 in Q1 journals), 6 as first author, and currently serve as a graduate course instructor at Dongguk University and peer reviewer for seven international journals, including the International Journal of Impact Engineering, Thin-Walled Structures, and Journal of Mechanical Science and Technology.

Summary

Total Journal Publications	18	Q1 Journal Publications	16
First-Authored Publications	6	Corresponding-Author Publications	6
Conference Presentations	4	Peer Review Roles	7
Total Citations	369	h-index	8

Education

Aug 2019 –
Dec 2024

Ph.D. in Mechanical Engineering
University of Manchester, Manchester, UK
Department of Mechanical, Aerospace and Civil Engineering, School of Engineering
Thesis: *Determination of dynamic flow stress equation based on discrete material data*
Supervisor: Prof. Qingming Li (Q.M. Li), Qingming.li@manchester.ac.uk

Sep 2015 –
Jul 2017

M.E. in Aeronautical and Astronautical Science and Technology
Harbin Institute of Technology (HIT), Harbin, China
School of Astronautics
Thesis: *Experimental investigation on the impact resistance of polymer-aluminium layered plates*
Supervisor: Prof. Wei Zhang, zhdawei@hit.edu.cn

Sep 2010 –
Jul 2014

B.E. in Aircraft Manufacturing Engineering
Civil Aviation University of China (CAUC), Tianjin, China
College of Aeronautical Engineering
Coursework: Mechanics of Materials, Aircraft Fault Diagnosis Technology, Engineering Materials, Aircraft Flight Mechanics, Aircraft Manufacturing Technology

Academic & Professional Appointments

Sep 2025 –
Present

Visiting Research Professor

	Dept. of Mechanical, Robotics and Energy Engineering, Dongguk University–Seoul Supervisor: Prof. Heung Soo Kim Prognostics and Health Management of Lithium-Ion Batteries Based on Neuromorphic Sensor and Artificial Intelligence Framework
Oct 2025 – Present	Postdoctoral Researcher InnoCORE PRISM-AI Center, KAIST, South Korea Supervisor: Prof. Seunghwa Ryu Prognostics and Health Management of Lithium-Ion Batteries Based on Neuromorphic Sensor and Artificial Intelligence Framework
May 2025 – Aug 2025	Research Associate (Part-time) Zhejiang University, China Supervisor: Prof. Guoxing Lu
Oct 2020 – Apr 2024	Research Associate University of Manchester, UK Supervisor: Prof. Qingming Li (Q.M. Li)

Teaching Experience

Sep 2026 – Dec 2026	Instructor (Graduate Course) Dongguk University, “Fundamentals of Prognostics and Health Management” (MEC6008-01) - Sole instructor for a graduate course on the fundamentals of prognostics and health management (PHM) for mechanical systems and components (bearings, gearboxes, rotor systems), covering the Industry 4.0 context, physics-based, data-driven, and hybrid PHM strategies, data preprocessing, and descriptive statistics. - Guided students through a semester-long project applying machine learning and deep learning methods to real PHM diagnostic and prognostic problems.
Mar 2026 – Jun 2026	Instructor (Graduate Course) Dongguk University, “Machine Learning and Its Engineering Application” - Sole instructor for a graduate-level course covering CNN, RNN, LSTM, and Transformer architectures, with emphasis on their application to mechanical and PHM systems; students implemented models hands-on for data-driven diagnosis and prediction problems. - Mentored a team of students competing in the 2nd KSPHM-KIMM Mechanical Data Challenge 2026, a national mechanical data-driven diagnostics competition (concluded June 25, 2026).
Sep 2019 – Dec 2023	Graduate Teaching Assistant University of Manchester, UK - Supported students completing the course Advanced Modelling & Simulation - Supported students completing the course Composites & Polymers - Demonstrated the operating procedure of column buckling experiments - Instructed students completing ANSYS numerical simulation coursework

Research Experience

My research spans two connected streams built on a common foundation of experimental impact mechanics: the impact safety and prognostics & health management (PHM) of lithium-ion batteries, and a data-driven framework for the dynamic constitutive modeling of metals.

2023 – Present	Lithium-Ion Battery Impact Safety and Prognostics & Health Management (PHM) Leading a battery PHM research group at the Dongguk iPHM Lab, developing neuromorphic-
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sensor- and AI-based frameworks for lithium-ion battery health monitoring.

- Co-supervised, as corresponding author, experimental characterization of lithium-ion battery failure modes and failure-energy thresholds under extreme impact, and multiphysics finite-element characterization of critical failure impact on battery pack stability. *Journal of Energy Storage* 150 (2026) 120037; *Engineering Failure Analysis* 196 (2026) 111084.
- Co-supervised, as corresponding author, a deep-learning framework using acoustic emission signals for real-time damage assessment of lithium-ion batteries under dynamic impact (CAUC-funded Graduate Research Project). *Journal of Power Sources* 662 (2026) 238737.
- Authored an invited review of the battery PHM field, synthesizing current challenges and future perspectives. *Journal of Mechanical Science and Technology* 40(4) (2026) 2405–2415.
- Drafted the rationale, research content, and literature review for a National Natural Science Foundation of China (NSFC) grant proposal on multiscale, multiphysics-coupled failure mechanisms of high-energy-density batteries under dynamic loading.

2019 –
2027

Data-Driven Constitutive Modeling of Dynamic Flow Stress

- Verified when decoupled, Johnson-Cook-type flow stress equations are mathematically legitimate for a given material, using the rank-1 energy ratio of an SVD decomposition of two-way flow stress data as a quantitative criterion. *International Journal of Impact Engineering* 173 (2023) 104453.
- Determined dynamic flow stress equations directly from discrete, non-ideal SHPB data by relaxing the constant-strain-rate requirement of classical reduction protocols, resolving the strain-rate dependence (Part 1) and extending to the full three-dimensional strain, strain-rate and temperature surface via CP tensor decomposition (Part 2). *International Journal of Impact Engineering* 206 (2025) 105403 and 105432.
- Developed TEDI-FS (Tensor-Decomposition-based Extrapolation and Dynamic Identification of Flow Stress), a framework that validly extrapolates an SHPB-calibrated flow stress model to very-high strain rates ($\sim 10^5 \text{ s}^{-1}$), inversely identified and validated against coupled Taylor-Hopkinson impact experiments and explicit finite element simulation. *International Journal of Impact Engineering* 219 (2027) 105863.
- The methodology combines SVD/CP tensor decomposition with artificial neural networks and applies beyond the Johnson-Cook equation to any factorized constitutive model, built on an explicit isotropic-metal (J2 plasticity) assumption stated in the underlying Ph.D. thesis.

2015 –
2026

Impact Behavior of Composite and Sandwich Structures

- Co-supervised, as corresponding author, ANN surrogate models for the ballistic resistance and residual-velocity prediction of finite-thickness metallic plates under impact. *Thin-Walled Structures* 203 (2024) 112161; *Thin-Walled Structures* 216 Part B (2025) 113685.
- Co-supervised, as corresponding author, a data-driven decoupled hardening model explaining non-uniform bulging in thin-sheet forming, using a machine-learning-assisted non-associated Hill48 framework in which anisotropic parameters evolve with accumulated plastic strain, validated against Erichsen cupping tests on Al5083-H112 sheets. *Journal of Materials Engineering and Performance* (2026).
- Contributed constitutive modeling and simulation analysis to the multiscale modeling of impact behavior and mesoscale damage in 3D angle-interlock woven composites. *International Journal of Mechanical Sciences* 276 (2024) 109382; *Composite Structures* 382 (2026) 120111.
- During master's research at HIT, designed and executed experiments, including DIC-based penetration testing, and finite element simulations for the dynamic response of sandwich structures and polymer-aluminum laminated plates under impulsive and ballistic loading; first author on the resulting *International Journal of Impact Engineering* 113 (2018) paper, co-author on three further *International Journal of Impact Engineering* papers (2016 to 2017), and one AIP Conference Proceedings paper.

Student Supervision

Master's students, co-supervised with Dr. Yunfei Deng:

2022 – 2025	Xiaoyue Yang • Impact mechanics; ANN surrogate models for the impact resistance of finite-thickness plates. • Co-authored: <i>Thin-Walled Structures</i> 203 (2024) 112161; <i>Thin-Walled Structures</i> 216 Part B (2025) 113685; <i>Journal of Materials Engineering and Performance</i> (2026). → Now: Ph.D. Candidate, Hunan University
2023 – 2026	Han Zheng • Mechanical safety of lithium-ion batteries: failure modes under impact and multiphysics response at the pack level. • Co-authored: <i>Journal of Energy Storage</i> 150 (2026) 120037; <i>Engineering Failure Analysis</i> 196 (2026) 111084. → Now: Ph.D. Student, Sun Yat-sen University
2023 – 2026	Jiangtao Li • Deep learning for real-time damage assessment of lithium-ion batteries under dynamic impact, with acoustic emission monitoring. • Co-authored: <i>Journal of Power Sources</i> 662 (2026) 238737. → Now: Ph.D. Candidate, Beijing Institute of Technology
2023 – 2026	Yixu Lv • Neural network surrogate models for residual velocity prediction in finite-thickness plates. • Co-authored: <i>Thin-Walled Structures</i> 216 Part B (2025) 113685. → Now: Materials Engineer, AEROFUGIA

Undergraduate & postgraduate thesis supervision, University of Manchester:

- Instructed undergraduate student thesis: Machine learning based dynamic constitutive models (10/2023 – 04/2024)
- Instructed postgraduate (master's degree) student thesis: Signal analysis of damaged space structure by impact (08/2020 – 07/2021)
- Instructed undergraduate student thesis: Structural health monitoring of space structures under impacts (10/2020 – 03/2021)

Academic Awards and Achievements

Jun 2026	Invited Speaker PHM Korea 2026 Annual Conference, Busan, Republic of Korea
Dec 2025	Session Chair KSME 3rd Emerging Technologies in Mechanical Engineering (ETME 2025), Kangwon Land, Republic of Korea
May 2022	UK Henry Lester Trust Grant University of Manchester, UK
May 2021	UK Henry Lester Trust Grant University of Manchester, UK

Journal Publications (SCIE-indexed: 18, Q1: 16)

2027

1. **X. Huang**, Q.M. Li*, Tensor-decomposition-based inverse characterization of flow stress for very-high strain-rate impact, *International Journal of Impact Engineering*, 219 (2027) 105863. IF: 6.7, Q1

2026

1. Y. Deng, W. Pu, X. Yang, **X. Huang***, A Data-Driven Decoupled Hardening Model: Uncovering the Fundamental Mechanism of Non-uniform Bulging in Thin Sheets, *Journal of Materials Engineering and Performance* (2026). IF: 2.6, Q3
2. Y. Deng, H. Zheng, T. Zhang*, **X. Huang***, Characterizing critical failure impact on LIB pack stability via multiphysics response, *Engineering Failure Analysis*, 196 (2026) 111084. IF: 7.0, Q1
3. **X. Huang**, H.S. Kim*, Advancing battery prognostics and health management: Challenges and future perspectives, *Journal of Mechanical Science and Technology*, 40(4) (2026) 2405–2415. IF: 2.2, Q3
4. H. Wei, **X. Huang**, X. Jiang, W. Xie, Y. Li, G. Zhao, W. Zhang*, Mesoscale damage modeling of 3D woven composites under quasi-static loads: Incorporating plasticity and fiber misalignment, *Composite Structures*, 382 (2026) 120111. IF: 7.8, Q1
5. Y. Deng, H. Zheng, **X. Huang***, Failure modes and failure energy threshold of lithium-ion batteries under extreme impact, *Journal of Energy Storage*, 150 (2026) 120037. IF: 10.7, Q1
6. Y. Deng, J. Li, **X. Huang***, Deep learning-based real-time damage assessment of lithium-ion batteries under dynamic impact, *Journal of Power Sources*, 662 (2026) 238737. IF: 8.4, Q1

2025

1. **X. Huang**, Q.M. Li*, Determination of dynamic flow stress equation based on discrete experimental data: Part 1 Methodology and the dependence of dynamic flow stress on strain-rate, *International Journal of Impact Engineering*, 206 (2025) 105403. IF: 6.7, Q1
2. **X. Huang**, Q.M. Li*, Determination of dynamic flow stress equation based on discrete experimental data: Part 2 Dynamic flow stress depending on strain, strain-rate and temperature, *International Journal of Impact Engineering*, 206 (2025) 105432. IF: 6.7, Q1
3. Y. Deng, Y. Lv, X. Yang, C. Du, **X. Huang***, Determination of residual velocity model of finite thickness material based on Artificial Neural Network, *Thin-Walled Structures*, 216 Part B (2025) 113685. IF: 8.3, Q1

2024

1. H. Wei, **X. Huang**, W. Xie, X. Jiang, G. Zhao, W. Zhang*, Multiscale modeling for the impact behavior of 3D angle-interlock woven composites, *International Journal of Mechanical Sciences*, 276 (2024) 109382. IF: 11.4, Q1
2. Y. Deng, X. Yang, **X. Huang***, Determination of ballistic resistance model of finite thickness material based on Artificial Neural Network, *Thin-Walled Structures*, 203 (2024) 112161. IF: 8.3, Q1

2023

1. **X. Huang**, Q.M. Li*, The legitimacy of decoupled dynamic flow stress equations and their representation based on discrete experimental data, *International Journal of Impact Engineering*, 173 (2023) 104453. IF: 6.7, Q1

2022

1. Y. Deng, H. Wu, Y. Zhang*, **X. Huang**, X. Xiao, Y. Lv, Experimental and numerical study on the ballistic resistance of 6061-T651 aluminum alloy thin plates struck by different nose shapes of projectiles, *International Journal of Impact Engineering*, 160 (2022) 104083. IF: 6.7, Q1

2018

1. **X. Huang**, W. Zhang*, Y. Deng, X. Jiang, Experimental investigation on the ballistic resistance of polymer-aluminum laminated plates, *International Journal of Impact Engineering*, 113 (2018) 212–221. IF: 6.7, Q1

2016–2017

1. N. Ye, W. Zhang*, D. Li, W. Huang, W. Xie, **X. Huang**, X. Jiang, Dynamic response and failure of sandwich plates with PVC foam core subjected to impulsive loading, *International Journal of Impact Engineering*, 109 (2017) 121–130. IF: 6.7, Q1
2. W. Huang, W. Zhang*, **X. Huang**, X. Jiang, Y. Li, L. Zhang, Dynamic response of aluminum corrugated sandwich subjected to underwater impulsive loading: Experiment and numerical modeling, *International Journal of Impact Engineering*, 109 (2017) 78–91. IF: 6.7, Q1
3. W. Huang*, B. Jia, W. Zhang*, **X. Huang**, D. Li, P. Ren, Dynamic failure of clamped metallic circular plates subjected to underwater impulsive loads, *International Journal of Impact Engineering*, 94 (2016) 96–108. IF: 6.7, Q1

Conferences

Presentations

1. **X. Huang** (Invited Speaker), H.S. Kim*, “Impact-Induced Damage Assessment and Remaining Useful Life Prediction of Lithium-Ion Cells Using Acoustic Emission Features,” *PHM Korea 2026 Annual Conference*, Busan, Republic of Korea, Jun 2026
2. **X. Huang** (Oral Presentation), H.S. Kim*, “Acoustic Emission Characteristics of Lithium-Ion Cells under Mechanical Impact,” *2026 Spring Conference of the Korean Society for Noise and Vibration Engineering (KSNVE)*, Gangwon-do, Republic of Korea, May 2026
3. **X. Huang** (Oral Presentation), H.S. Kim*, “Impact-Induced Pulse Power Capability Degradation in Lithium-Ion Cells: Experimental Characterization and Structural Response Analysis,” *2026 Spring Conference of the CAE and Applied Mechanics Division, KSME*, Yeosu-si, Republic of Korea, May 2026
4. **X. Huang** (Oral Presentation), H.S. Kim*, “Physics-Informed Probabilistic Machine Learning for Battery Degradation Diagnostics under Dynamic Mechanical and Electrochemical Conditions,” *KSME 3rd Emerging Technologies in Mechanical Engineering (ETME 2025)*, Kangwon Land, Republic of Korea, Dec 2025

Published Proceedings

1. W. Zhang*, **X. Huang**, Y. Qi, D. Li, J. Tao, W. Huang, Experimental investigation on ballistic stability of high-speed projectile in sand, *AIP Conference Proceedings* 1793, 060031 (2017)

Technical Skills

- **Simulation Software:** ABAQUS, LS-DYNA: explicit modelling of dynamic impact with user-defined subroutines (VUMAT/UMAT)
- **Programming Languages:** C, Python, MATLAB, Fortran
- **Deep Learning Frameworks:** PyTorch, TensorFlow
- **CAD / Geometric Modelling:** CATIA

Peer Review Board Membership (7 journals)

- *International Journal of Impact Engineering* (Elsevier)
- *Scientific Reports* (Springer Nature)
- *Defence Technology* (Elsevier)
- *Measurement* (Elsevier)
- *Thin-Walled Structures* (Elsevier)
- *Journal of Mechanical Science and Technology* (Springer)
- *Journal of Advanced Energy Research*

References

Prof. Qingming Li (Q.M. Li), Ph.D. Supervisor

- Professor, School of Engineering, University of Manchester, UK
- Email: Qingming.li@manchester.ac.uk

Prof. Heung Soo Kim, Current Supervisor

- Director, BK21 AIMS Center; Director, DGU Global Smart Factory Center
- Dept. of Mechanical, Robotics and Energy Engineering, Dongguk University, South Korea
- Email: heungsoo@dgu.edu

Prof. Seunghwa Ryu, Current Supervisor

- InnoCORE PRISM-AI Center, Korea Advanced Institute of Science and Technology (KAIST), South Korea
- Email: ryush@kaist.ac.kr